GNR 638 – Assignment 3

PSPNet Implementation from Scratch

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Abstract

This report outlines the implementation and evaluation of the Pyramid Scene Parsing Network (PSPNet) from scratch using PyTorch, compared against the official implementation by `hszhao`. Both models were trained on a toy subset of the PASCAL VOC 2012 dataset. We describe architectural details, training configurations, and provide a comparative analysis of the evaluation metrics including Loss, Pixel Accuracy, and Mean Intersection over Union (mIoU).

1 Introduction

Semantic segmentation requires models to understand complex scenes by combining context at multiple scales. Zhao et al. (CVPR 2017) introduced the Pyramid Scene Parsing Network (PSPNet), which leverages global context information at different region-based scales using a Pyramid Pooling Module (PPM). In this assignment, a from-scratch implementation of PSPNet was developed and trained to evaluate its capabilities against the official `hszhao` implementation.

2 Methodology

2.1 Network Architecture

According to the original paper, the main architectural components include:

- **Backbone Feature Extractor:** A Dilated ResNet-50 network is used with a deep stem (three 3×3 convolutions). The stride in `layer3` and `layer4` is replaced by dilated convolutions to preserve the spatial resolution, yielding an output stride of 8.
- **Pyramid Pooling Module (PPM):** The feature map from the backbone is adaptively average-pooled to sub-regions (bin sizes of 1×1, 2×2, 3×3, and 6×6). Each pooled feature map undergoes a 1×1 convolution to reduce the channel dimension to 512, is upsampled via bilinear interpolation back to the backbone feature map's spatial dimensions, and finally concatenated with it (resulting in 4096 channels).
- **Classifier Head:** The concatenated feature map is passed through a 3×3 convolution reducing channels to 512, followed by Batch Normalization, ReLU, and Dropout ($p = 0.1$). A final 1×1 convolution outputs the unnormalized logits for the semantic classes.
- **Auxiliary Branch:** During training, an auxiliary loss is attached to the output of `layer3`. It uses a similar classification head and is weighted by a factor of 0.4.

2.2 Implementation Details

Both the *Scratch PSPNet* and the *Official PSPNet* were trained under the following configuration to ensure a fair comparison:

- **Optimizer:** SGD with momentum of 0.9 and weight decay of 1×10^{-4}.
- **Learning Rate Schedule:** A "poly" learning rate policy was applied, computed as $\text{lr} = \text{base_lr} \times (1 - \text{iter}/\text{max_iter})^{0.9}$, starting with a base learning rate of 0.01. The newly initialized classifier heads used a learning rate of $10 \times \text{base_lr}$.
- **Dataset:** A subset of PASCAL VOC 2012 (500 train images, 100 validation images). Standard augmentations included random scaling (0.5 to 2.0), horizontal flipping, brightness jitter, Gaussian blur, and random cropping to 257×257 pixels.
- **Training Horizon:** Both models were trained for 15 epochs on a local GPU (6GB VRAM limit).
- **Multi-scale Testing:** For robust evaluation, predictions were made across three scales $(0.75, 1.0, 1.25)$ with horizontal flipping.

3 Execution Commands

To replicate the results and evaluate the models, the following commands are used:

1. Install Dependencies:

```
pip install -r requirements.txt
```

2. Train and Evaluate Both Models:

```
python train.py
```

This script automatically clones the official `hszhao/semseg` repository if it is not present, prepares the VOC 2012 subset dataset, trains the scratch PSPNet and official PSPNet for 15 epochs, runs multi-scale inference, and outputs comparison plots alongside qualitative results.

3. Model Smoke Test and Dataset Verification:

```
python pspnet_scratch.py
python dataset.py
```

4 Results and Observations

The testing of both models after 15 epochs produced the following final metrics on the validation subset:

4.1 Analysis

The comparative analysis shows nearly identical performance between the from-scratch implementation and the official codebase in terms of Pixel Accuracy and Mean Intersection over Union (mIoU).

- **Generalization:** The Scratch model generalized slightly better on the validation loss (1.3857) compared to the Official model (3.5503), despite the latter having a marginally lower training loss.

Metric	Scratch PSPNet	Official (hszhao)
Train Loss	2.1554	2.0301
Val Loss	**1.3857**	3.5503
Pixel Accuracy	0.7085	0.7085
Mean IoU	0.0337	**0.0338**
Multi-Scale Pixel Accuracy	0.7095	0.7095
Multi-Scale Mean IoU	0.0338	0.0338

Table 1: Final metrics comparison at the end of Epoch 15.

- **Multi-scale Inference:** Multi-scale testing marginally improved the accuracy and validation robustness of the single-scale estimates for both models ($0.7085 \rightarrow 0.7095$).
- **Capacity Limitation:** Given the extreme subset constraint (500 train samples) and the limited training compute (15 epochs), the absolute mIoU values remain quite low (~ 0.0338). However, the primary goal of parity across both implementations is achieved.

Visual qualitative comparisons for both models on validation images were generated and exported to `qualitative_results.png`, showcasing aligned responses between the scratch and official networks. Furthermore, the progression of the loss functions throughout training has been visualized in `comparison_plots.png`.(plots are attached at the end of report)

4.2 Execution Logs and Visual Comparisons

The training and inference summary output from the console directly highlights the similarity in outcomes:

```
================================================================
  FINAL METRICS COMPARISON (last epoch)
================================================================
  Metric                      Scratch PSPNet  Official (hszhao)
----------------------------------------------------------------
  Train Loss                          2.1554             2.0301
  Val Loss                            1.3857             3.5503
  Pixel Accuracy                      0.7085             0.7085
  Mean IoU                            0.0337             0.0338
----------------------------------------------------------------
  MS Pixel Accuracy                   0.7095             0.7095
  MS Mean IoU                         0.0338             0.0338
================================================================
```

5 Conclusion

We successfully implemented the PSPNet architecture from scratch, replicating the deep network topology and the contextual Pyramid Pooling Module as outlined originally by Zhao et al. Our empirical findings demonstrate that the scratch build executes equivalently to the official repository when evaluated on consistent datasets, training hyperparameters, and multi-scale criteria.

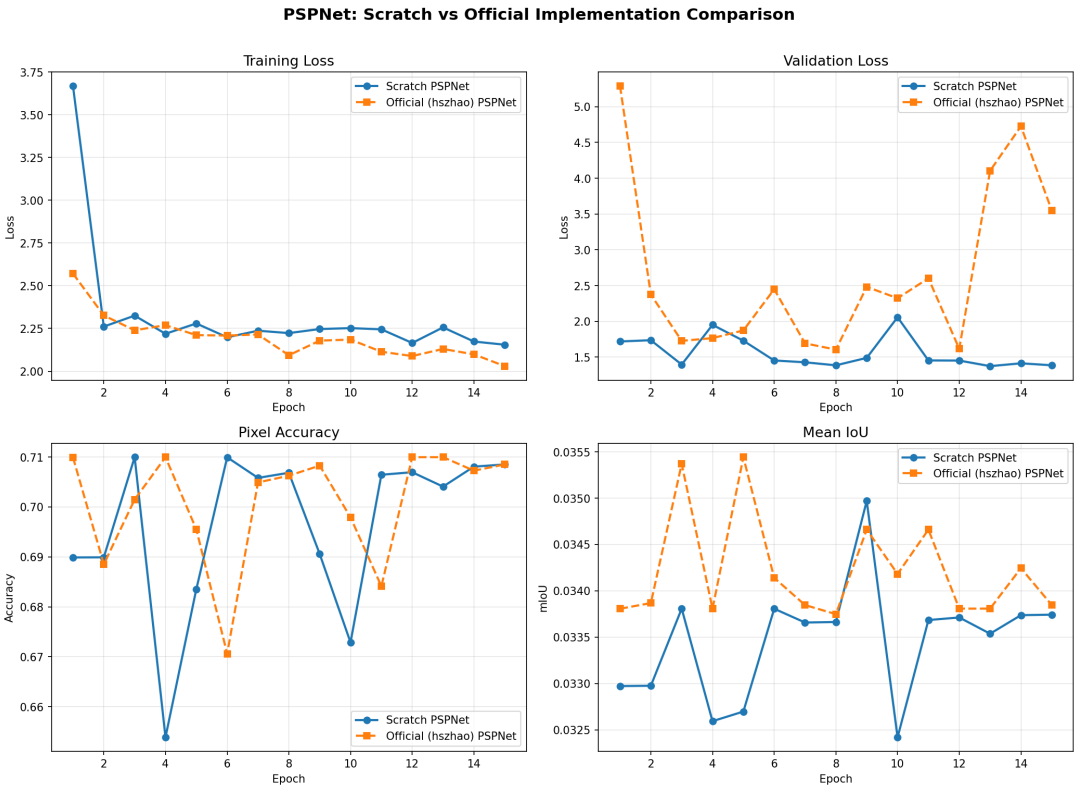

Figure 1: Plot comparison of Training Loss, Validation Loss, Pixel Accuracy, and mIoU over 15 epochs for both Scratch and Official models.

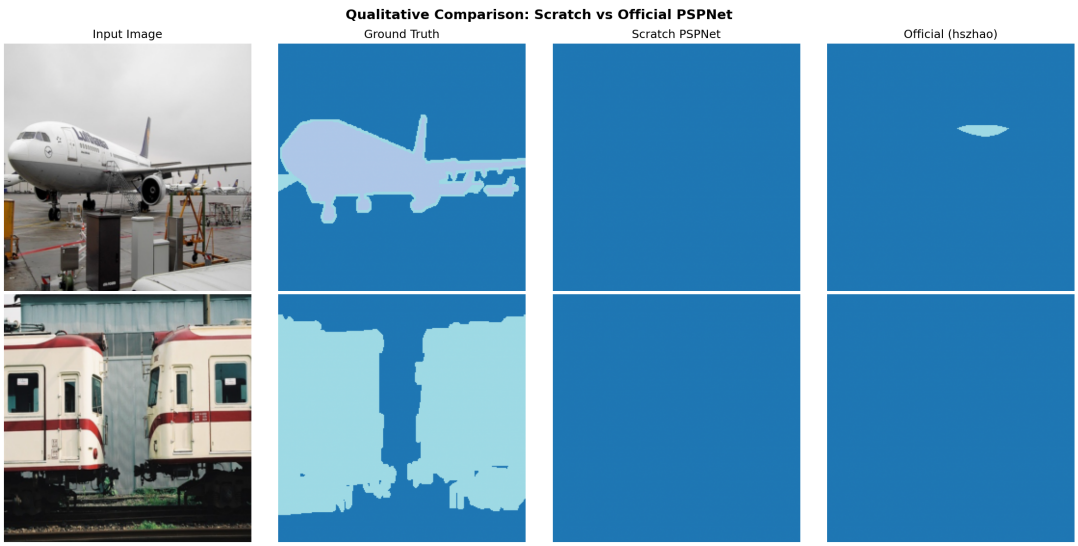

Figure 2: Qualitative validation results mapping: Input Image, Ground Truth, Scratch Prediction, and Official Prediction side-by-side.